Exercise 1

3, 4, 4, 5, 5, 5, 6, 6, 7, 7, 8, 8, 8, 8, 8, 9, 10, 10, 11, 13

a) $$\text{mean} = \frac{3+4+4+5+5+5+6+6+7+7+8+8+8+8+8+9+10+10+11+13}{20}$$

$$= \frac{145}{20}$$

$$= 7.25$$

b) $$\text{position} = \frac{n+1}{2}$$

$$= \frac{20+1}{2}$$

$$= 10.5\text{th}.$$

$$\text{median} = \frac{10\text{th} + 11\text{th}}{2}$$

$$= \frac{7+8}{2}$$

$$= 7.5$$

c) mode = 8.

Exercise 2

a) mean $= \frac{\Sigma fx}{fx} = \frac{4(2001)+5(2002)+4(2003)+52(2004)+10(2005)+5(2006)+3(2007)+5(2008)}{4+5+4+52+10+5+3+5}$

$= \frac{176375}{88}$

$= 2004.2614.$

position of median $= \frac{88+1}{2}$

$= 44.5$th.

median $= \frac{44\text{th} + 45\text{th}}{2}$

$= \frac{2004+2004}{2}$

$= 2004.$

mode $= 2004.$

b) They could use the mean to argue for improvements. It shows the average of accidents.

c) They could use the mode for argument.
The outlier 52 in 2004 shows that accidents occur are infrequent.

Exercise 3

Length (mm)	Frequency	midpoint	cf
149.5 – 154.5	5	152	5
154.5 – 159.5	2	157	7
159.5 – 164.5	6	162	13
164.5 – 169.5	8	167	21
169.5 – 174.5	9	172	30
174.5 – 179.5	11	177	41
179.5 – 184.5	6	182	47
184.5 – 189.5	3	187	50

a)
$$\text{mean} = \frac{5(152)+2(157)+6(162)+8(167)+9(172)+11(177)+6(182)+3(187)}{5+2+6+8+9+11+6+3}$$
$$= \frac{8530}{50}$$
$$= 170.6$$

b) median position $= \frac{50}{2} = 25\text{th}.$

$$\text{median} = L + \left[\frac{\frac{n}{2} - F}{f}\right] \times C$$
$$= 169.5 + \left[\frac{\frac{50}{2} - 21}{9}\right] \times 5$$
$$= 172.$$

c)
$$\text{mode} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h$$
$$= 174.5 + \left[\frac{11 - 9}{2(11) - 9 - 6}\right] \times 5$$
$$= 175.929.$$

Exercise 4.

Age	Students	Faculty/Staff
0.5 – 3.5	23	30
3.5– 6.5	33	47
6.5 – 9.5	63	36
9.5 – 12.5	68	30
12.5 – 15.5	19	8
15.5 – 18.5	10	0
18.5 – 21.5	1	0
21.5 – 24.5	0	1
	total 217	total 152.

midpoint	cf (students)	cf(staff)
2	23	30
5	56	77
8	119	113
11	187	143
14	206	151
17	216	151
20	217	151
23	217	152

$$\text{mean (students)} = \frac{23(2)+33(5)+63(8)+68(11)+19(14)+10(17)+1(20)+0(23)}{23+33+63+68+19+10+1+0}$$

$$= \frac{1919}{217}$$

$$= 8.843.$$

$$\text{mean(staff)} = \frac{30(2)+47(5)+36(8)+30(11)+8(14)+0(17)+0(20)+1(23)}{30+47+36+30+8+0+0+1}$$

$$= \frac{1048}{152}$$

$$= 6.8947.$$

$$\text{median(student)} = L + \left[\frac{\frac{n}{2} - F}{f}\right] \times c$$

$$= 6.5 + \left[\frac{\frac{217}{2} - 56}{63}\right] \times 3$$

$$= 9$$

$$\text{median position} = \frac{217+1}{2}$$

$$= 109\text{th}.$$

$$\text{median (staff)} = L + \left[\frac{\frac{n}{2} - F}{f}\right] \times c$$

$$= 3.5 + \left[\frac{\frac{152}{2} - 30}{47}\right] \times 3$$

$$= 6.4362$$

$$\text{median position} = \frac{152}{2}$$

$$= 76\text{th}.$$

$$\text{mode (student)} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h$$

$$= 9.5 + \left[\frac{68 - 63}{2(68) - 63 - 19)}\right] \times 3$$

$$= 9.778.$$

$$\text{mode(staff)} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h$$

$$= 3.5 + \left[\frac{47 - 30}{2(47) - 30 - 36}\right] \times 3$$

$$= 5.321$$

Exercise 5.

a) i) $Y[k] = 0.8143$

$$\text{percentile of value } 0.8143 = \frac{\text{no of values} < 0.8143}{\text{total no of values}}(100) = \frac{8}{18}(100)$$
$$= 44.44$$

$\therefore$ the value 0.8143 is at the 45th percentile.

ii) $Y[k] = 0.8062$

$$\text{percentile of value } 0.8062 = \frac{\text{no of values} < 0.8062}{\text{total no of values}}(100) = \frac{2}{18}(100)$$
$$= 11.11$$

$\therefore$ the value 0.8062 is at the 12th percentile.

bi) $i = \frac{80}{100}(18)$
$= 14.4\text{th}$
$\approx 15\text{th}$

$P_{80} = 0.8161$

ii) $i = \frac{3}{4}(18)$
$= 13.5\text{th}$
$\approx 14\text{th}$

$Q_3 = 0.8161$

iii) $i = \frac{33}{100}(18)$
$= 5.94$
$\approx 6\text{th}.$

$P_{33} = 0.8110$

iv) $i = \frac{1}{4}(18)$
$= 4.5$
$\approx 5\text{th}$

$Q_1 = 0.8079.$

Exercise 6

34 36 36 38 38 39 39 40 40 41 41 41 41 42 42 45 45 49 56

$$\text{range} = 56 - 34$$
$$= 22.$$

$$\text{mean} = \frac{\Sigma fx}{fx} = \frac{783}{19}$$
$$= 41.21$$

$$\text{variance} = \frac{\Sigma fx^2 - \frac{(\Sigma fx)^2}{n}}{n-1}$$

$$= \frac{34^2 + 2(36)^2 + 2(38)^2 + 2(39)^2 + 2(40)^2 + 4(41)^2 + 2(42)^2 + 2(45)^2 + 49^2 + 56^2 - \frac{(783)^2}{19}}{19-1}$$

$$= \frac{32717 - 32267.84}{18}$$

$$= 24.95$$

$$\text{standard deviation} = \sqrt{24.95}$$
$$= 4.9950$$

Exercise 7.

$$\text{mean}, \bar{x} = \frac{\sum x}{n}$$

$$= \frac{10 + 50 + 30 + 20 + 10 + 20 + 70 + 30}{8}$$

$$= \frac{240}{8}$$

$$= 30$$

x	$(x-30)$	$(x-30)^2$	$(x-30)^3$
10	-20	400	-8000
50	20	400	8000
30	0	0	0
20	-10	100	-1000
10	-20	400	-8000
20	-10	100	-1000
70	40	1600	64000
30	0	0	0
—	—	—	—
240	9	3000	54000

$$\text{standard deviation} = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{3000}{8-1}}$$

$$= 20.70$$

$$\text{sample skewness} = \frac{\sum (x-\bar{x})^3}{(n-1) \cdot s^3}$$

$$= \frac{54000}{7 \cdot (20.70)^3}$$

$$= 0.8697.$$